# Flight events from ADS-B: fixing what OPDI publishes, and reaching the GANP KPIs

**Version 1**

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OPDI has published flight events since v0.0.2 without a single test or a single comparison against reference data. This study audits the four detectors behind them, finds nine defects — one of which silently suppresses every take-off and landing at any aerodrome above roughly 200 ft — fixes them, and adds four milestone families chosen because they are what the ICAO GANP key performance indicators actually require. It reports what has been measured and, equally, what has not.

## Table of contents



## 1 What this study is

Two things prompted it.

**The events were never checked.** OPDI's flight-event table is the least-examined thing the project publishes. Departure and destination aerodrome detection has been through seven successive studies and a full rewrite; the event detectors had no tests at all, and no number in them had ever been compared against anything.

**The event vocabulary did not reach the indicators.** The stated goal is to compute as many ICAO GANP key performance indicators as ADS-B allows. Mapping the published KPI list against what OPDI emitted showed the gap concentrated in four milestone families — and that ICAO itself names ADS-B data providers as a feed for two indicators OPDI could not compute at all.

## 2 What was wrong

Nine defects, found by reading the detectors. None of them fails. Each produces fewer events, or events displaced in one direction — the failure mode that survives an eyeball check of the totals.

| | Defect | Consequence |
|---|---|---|
| D1 | Ground membership measured against **uncorrected pressure altitude** | The membership reaches zero at 200 ft *pressure* altitude, so at any aerodrome above roughly 200 ft AMSL — or anywhere on a low-pressure day — no sample is ever classified as on-ground. Since take-off requires a preceding ground sample and landing requires a ground sample, **neither event has ever been emitted for those flights.** The loss is biased against high-elevation aerodromes. |
| D2 | The reference implementation's 60-second majority smoothing was **not ported** | Phases were decided per state vector at 5-second spacing with no temporal aggregation, so one misclassified sample injects a spurious level segment or destroys a take-off. |
| D3 | Cruise speed membership centred on 600 kt | A turboprop at 250 kt scores 0.002, so it never reaches cruise and gets no top-of-climb or top-of-descent. **Inherited from OpenAP, not introduced here** — left unchanged pending measurement. |
| D4 | `least`/`greatest` skip nulls | A rule missing an input became a minimum over fewer terms, which can only *raise* its activation — so incomplete evidence out-competed complete evidence. |

| | Defect | Consequence |
|---|---|---|
| D5 | `first`/`last` inside a group-by | Take partition order, not time order, so a reported entry position was not guaranteed to be the position at the reported entry time. |
| D6 | Events read the **raw** track table | Trajectory cleaning reached the flight list and nothing else, so no published event has ever been derived from a cleaned trajectory. |
| D7 | Non-reproducible event identifiers; one version string for every type | The same event got a different identifier on every run, so two runs of a month could not be reconciled. |
| D8 | Flight-level crossings kept **only the first and last** per level, uninterpolated | A flight that levels off and re-climbs through a level lost every crossing in between — exactly the crossings a vertical-efficiency indicator is about — and every reported instant was biased by up to one sample interval in the same direction. |
| D9 | No tests | Not one detector was covered. |

D1 and D2 are the two that matter most, because they mean the published take-off, landing, top-of-climb and top-of-descent events are missing **non-randomly**.

## 3 What was built

Eight defects are fixed and the ninth (D3) is left in place deliberately, because changing a constant inherited from the reference implementation should follow a measurement rather than precede one. Four milestone families were added:

- **Threshold crossings**, rewritten. Every crossing rather than the first and last, with a hysteresis dead band so an aircraft cruising *at* a level emits none, and the instant interpolated to the exact threshold.
- **ASMA ring crossings** at 40 and 100 NM, built around the flight's own origin and destination — which is what the indicators mean, and what the reference data records.
- **ICAO level segments**, implementing the published algorithm rather than reusing the fuzzy-phase level events. The two are **not interchangeable**.
- **Runway identification with ATOT and ALDT**, and **off-block/on-block**.

Every change is reversible: a `legacy()` configuration reproduces published `events_v0.0.2` exactly, and the new algorithm carries a new version string.

## 4 Which KPIs this reaches

| KPI | Title | Reachable from ADS-B? |
|---|---|---|
| KPI02 | Taxi-out additional time | Yes |
| KPI05 | Actual en-route extension | Yes — ICAO names ADS-B as the feed |
| KPI08 | Additional time in terminal airspace | Yes |
| KPI10 | Airport peak throughput | Yes |
| KPI13 | Taxi-in additional time | Yes, coverage-limited |
| KPI15 | Flight time variability | Yes |
| KPI17 | Level-off during climb | Yes |
| KPI18 | Level capping during cruise | Yes (variant 2) |
| KPI19 | Level-off during descent | Yes |
| KPI01, KPI14 | Departure / arrival punctuality | Actual half only — the scheduled half needs a schedule feed |
| KPI11 | Airport throughput efficiency | Numerator only |
| Others | ATFM, capacity, flight plan, fuel, safety, noise | No |

Nine fully and three partially, from four milestone families.

## 5 Ground truth, and its ceiling

Reference data is EUROCONTROL APDF. It is in movement form — there is no literal take-off-time column, and `SRC_PHASE` decides what a row carries — and crucially it has **no aircraft address**, so it cannot be joined to ADS-B directly. The route runs through the flight table on an internal key.

That bridge is the ceiling on every coverage figure in this study: a milestone that cannot be reached is indistinguishable from one that was missed, unless the ceiling is known. So it is measured first.

Table 1

| Period | APDF movements | Reaching `icao24` | Share |
|---|---|---|---|
| 2024-06 | 1,161,115 | 1,087,091 | **93.62%** |
| 2025-06 | 1,224,742 | 1,154,338 | **94.25%** |

Both periods clear 93%, so the ceiling is high enough not to dominate any result. The residual loss is roughly three per cent of movements carrying no internal key, and a further two per cent whose key has no counterpart in the flight table.

Two traps in this join are worth recording, because both fail silently:

**The reference carries the commercial flight number, ADS-B carries the ATC call-sign.** The two disagree on 20.4% of rows — `LH416` against `DLH416` — and joining on the wrong one discards a fifth of the sample while looking exactly like a detection failure.

**Timestamps are tagged Europe/Paris despite being named UTC.** The instants are correct, so converting is right and stripping the zone is a two-hour error. Spark's session timezone is set to UTC in the local session builder and not in the distributed one, so the loader asserts it rather than assuming it.

## 6 What has *not* been measured

This is version 1, and the honest position is narrower than the work might suggest.

**The ladder has not been run.** The benchmark harness is built, tested and verified end to end on the ground-truth side, but the eleven-rung comparison from published `events_v0.0.2` to the shipped configuration requires many hours of serial cluster time and has not been executed. **No accuracy or coverage figure for any detector appears in this study**, including for the defects the authors are confident about. A defect demonstrated by a unit test is a defect demonstrated on a synthetic trajectory, not on traffic.

**The vertical family cannot be scored against data at all.** No source available to OPDI holds level-segment truth. The claim made for it is conformance to ICAO's published algorithm, tested on trajectories whose geometry is known by construction — which is checkable, where "accurate" is not.

**Touchdown and lift-off times are proxies.** They are the extremes of the runway-detection window. Their bias against the reference is exactly what the unrun ladder would report.

**A useful yardstick, for when it is run.** The reference data contains two independent EUROCONTROL derivations of the same ring crossing, and they differ by −9 s at the 10th percentile and +11 s at the 90th. That spread is the floor on what agreement can mean here, and is a better basis for a tolerance than a round number chosen for looking reasonable.

## 7 Breaking change for consumers

The new version does **not** re-emit `first-xing-fl*` and `last-xing-fl*`. Those types keep their meaning under the frozen `events_v0.0.2`, so published data stays reproducible, but consumers moving to the new version must migrate to the crossing sequence number carried in each event's `info`: the first crossing is sequence 1, the last is the maximum.

Consumers should also know that `level-start`/`level-end` and the new ICAO `level-off-*` family answer different questions and are not interchangeable, and that the event table grows materially — every crossing is now emitted where previously only two per level were.

## 8 Provenance

Table 2

| Output | Script | Git SHA | Produced |
|---|---|---|---|
| `bridge_2024.json` | `benchmarks/events_gt.py` | `b0ac011` | 2026-08-14 |
| `bridge_2025.json` | `benchmarks/events_gt.py` | `b0ac011` | 2026-08-14 |

Every figure above is produced by `benchmarks/regenerate_events.py`, which re-runs a job only when the source files that job depends on have changed. Outputs with no manifest entry are reported as unverified rather than shown as fact. The ladder outputs are absent from this table because they have not been produced.